

CS 524: Introduction to Optimization Fall 2025

Author: Gokulnath Sourirajan (Joint with Thilak Raj Murugan)

Sustainable Race Calendar Optimization for Formula One - Project Spec**1. What is the issue being addressed?**

Formula One (F1) aims to achieve net-zero carbon by 2030, but most of its emissions come from logistics. The current race calendar forces teams to travel large intercontinental distances repeatedly throughout a season. This project treats the scheduling as a routing optimization problem to design a data-driven race calendar that minimizes logistics-based carbon emissions by factoring in transport modes and realistic constraints, while maintaining operational feasibility.

2. Where does the data come from and how will it be obtained?

Three main datasets will be used. Geospatial data for circuit and airport coordinates (via Geopy/Google Maps API) and the Google Distance Matrix API will provide air and road distances for emission calculations. Climate data from TerraClimate will define feasible weather windows for each venue. Cultural data (e.g., Ramadan, Golden Week) will be sourced from official calendars to avoid scheduling conflicts. Together, these ensure the optimized calendar is both practical and environmentally efficient. ✓

3. What is the optimization problem underlying this project?

The objective is to identify a race order that minimizes total logistics-related carbon emissions while maintaining a practical season layout. Each venue has an associated location, feasible time window, and travel cost determined by distance and transport mode. The schedule must also satisfy several operational constraints, including fixed opening and closing races, limits on consecutive events (to control double and triple headers), and a mandatory mid-season break. Together, these ensure that the optimized calendar remains both environmentally efficient and logistically realistic.

4. What are the deliverables?

how is this modelled - return to base, etc?

The first deliverable is a baseline optimized calendar that minimizes logistics-based emissions while keeping the location of opening and closing races to be fixed as Australia and Abu Dhabi respectively. The second is an impact analysis showing how changes in scheduling flexibility (such as adjusting the opening or closing race location) could further reduce emissions. These will be presented with data visualizations to highlight the environmental benefits of optimized scheduling.

5. Other points for me to consider when evaluating

Emission estimates will use public data and standard conversion factors. The project will begin with a simplified problem and later expand to include multiple team bases for realistic analysis.